HridayMitra: warning a family 3 days before a heart-failure hospital visit, from a 90-second check-in at home

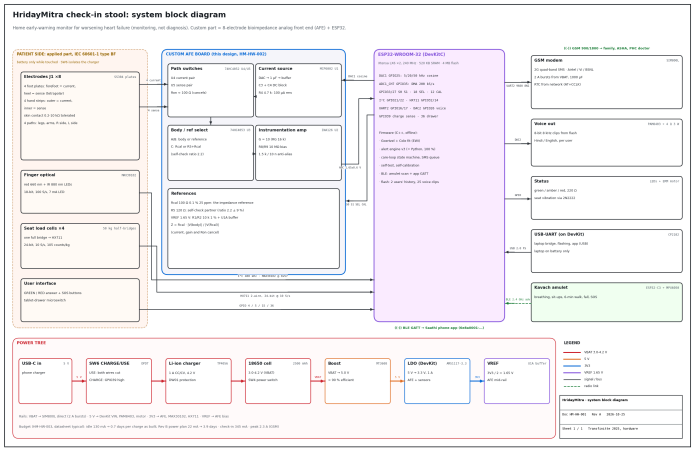
Transfinitte 2026 hardware track · Dare 2 Dream Ventures PS · **Monitoring, not diagnosis.** All patient data synthetic; no human data collected.

Problem

Fluid builds up silently for days before a heart-failure admission. Daily weighing alone catches about one event in three and floods families with false alarms; elders do not wear sticky patches; rural homes have power cuts, 2G and low literacy.

Solution: a check-in set for the patient's own chair + an amulet + a care loop

- **Weighing platform under the patient's own chair**: 4 load cells (weight), 4 stainless foot plates (heel + forefoot per foot). **Hand sensor bar**, held on the lap or **mounted on the chair's armrests / handles**: 4 hand strips, fingertip optical cradle (pulse, SpO_2), ✓ / ✗ / SOS buttons, speaker, status light. No chair modification; armless and arm chairs both work, never wheeled chairs.
- **Measurement**: weight, **8-electrode bioimpedance spectroscopy** at 5 / 20 / 50 kHz (legs, arms, both sides → chest fluid by segmental subtraction; water outside cells by a Cole fit), pulse + SpO_2, 3 spoken yes/no questions.
- **Kavach amulet** (ESP32-C3 + accelerometer, breakaway cord, no magnets): night breathing, sitting up at night (orthopnea), activity, Sunday 6-minute walk, falls, SOS → Bluetooth.
- **Offline engine on the ESP32**: learns each person's own normal, scores only the worsening direction, needs **two independent witnesses** before red, and says **why**. SMS care loop: family 1 → family 2 (30 min) → ASHA (2 h) → PHC doctor (6 h).
- **A product**: speaks the **doctor's own sick-day plan** in Hindi; coaching tips; two users with separate baselines; tablet-drawer switch; daily self-test with "needs service" SMS; Saathi phone app (offline, 10 languages, family / ASHA / doctor views).



System block diagram HM-HW-001: every bus labelled, patient-side (type BF) boundary, power tree.

User journey (90 s a day; one-page storyboard: *HridayMitra user journey.pdf*)

After morning tea Ramesh, 62, sits in his own chair on the platform, bare feet on the plates, hands on the bar; voice in Hindi; 3 questions; green light. In the storyboard, built on synthetic test patient P0189's real engine output: tablets missed days 32-34 → amber day 34 ("after missed medicine") → red day 41 with reasons by SMS → ASHA visit → the doctor's plan starts **4 days before** he would have needed hospital care.

Results (patient-level 70/30 split; thresholds frozen on the 140 training patients, then tested once)

	HridayMitra rules (on chip)	HridayMitra hybrid	Weight rule (SCALE-HF)	Logistic regression
Sensitivity, held-out (35 events)	66%	**86%**	40%	80%
Specificity (normal days with no red)	98.7%	97.9%	97.6%	-
Median lead before the hospital visit	3 days (amber: 4)	3 days	2 days	2 days
False alerts per patient-month (budget 0.21)	**0.06**	0.19	0.53	0.12
External 1,000-patient cohort: sensitivity / false alerts	72% / 0.06	**86% / 0.09**	31% / 0.38	81% / 0.08

Hybrid = the rules propose a red at a lower threshold, a logistic model trained only on dev patients must confirm; reasons stay the rules'; SpO_2 < 90% always red (86% on the stress cohort too). **Data**: 200 synthetic patients × 60 days (PS cohort) + 1,000 × 180 days (external), with salty meals, infection, missed pills, guests, wet feet and missing days. **Calibrated to real trials**: the published weight rules applied exactly as defined score on our external cohort like on real patients: FAST 20% caught (published 23%), 4.8 unexplained alerts / year (4.3); SCALE-HF 1 33% (35%), 4.8 / year (4.2). Open benchmark with scorer and splits: *benchmark/*.

Robustness: we tried to break it

Test	Result
Ablations (remove one sensor group)	no impedance 65%, no weight 66%, no amulet 54% (full engine 72%): every group earns its place
Stress: 50% missing days / guests / wet feet / power cuts / electrode ageing / everything at once	47% … 57% sensitivity; 10% guests: engine 0.10 false alerts vs weight rule **3.2**
Assumption sweep: each guess pushed low and high (52 new cohorts)	beats the weight rule in **52/52**, inside the false-alert budget in 52/52; sensitivity 30%–98%
Care-loop simulation (5,000 alerts per scenario)	clinical review before the visit: **92%** with the ladder vs 27% with one SMS; median 6 h; everything bad at once 68%
Heart rate on real ECG (PhysioNet BIDMC CHF, 15 records)	median error 0.15 bpm; 13 of 15 records within 0.5 bpm; 96% of 30-s windows within 5 bpm; fails on chf12 (very noisy)

Hardware and simulation proof (free tools: LTspice, KiCad, Wokwi, Python)

- **Design package**: block diagram HM-HW-001, 4-sheet schematic set HM-HW-002 (AFE, ESP32, power + isolation), power budget HM-HW-003, KiCad schematic (ERC 0, netlist checked against the path table and LTspice values).
- **Front end**: tetrapolar, ratiometric, $Z = Rcal \cdot |V_{body}| / |V_{Rcal}|$ (current, gain and switch resistance cancel). LTspice on the legs path: 578.4 Ω vs 578.5 Ω exact, 108 µA; every test-point waveform on the schematic sheet.
- **Monte Carlo, 100 boards from 5% parts**: 99% of readings within 1.3%, a +10% fluid change within 0.10 points; it changed the design (bias 1 MΩ → 10 MΩ).
- **Spectroscopy**: +10% fluid outside cells moves the index +3.0%; foot position / temperature moves the single-frequency reading +3.0% but the index +0.0%.
- **Patient safety (IEC 60601-1, type BF)**: every single fault ≥ 12.8× under the limit; the CHARGE / USE switch cuts charger leakage from 339 to 0.19 µA. **Self-calibrating** 0.1% reference (median error 0.16%); mains hum ≤ 0.16% with Goertzel vs 18% peak-to-peak.
- **Firmware proof**: C++ engine = Python on every test day; software-in-the-loop month with the real firmware: 41/41 checks (care loop, power cut, network outage, lifted foot, app link); Wokwi run and a recorded demo video.

Edge cases handled

Situation	What happens
Guest sits down	weight jump > 2.5 kg → nothing saved
Foot off a plate / toe ring	names the foot, re-measures; > 25% off → "no reading"
Power cut, no network	18650 battery; SMS queued, retried every 10 min
Missed check-in	11 am reminder to the family; light held
Infection vs fluid	pulse / breathing up, fluid steady → "maybe infection"
Oxygen below 90%	red at once, everyone messaged
Low literacy, hearing loss	voice, lights, ✓ / ✗ buttons, missed call = reply; seat vibration
Charger plugged in	refuses to measure (CHARGE sense)
Pacemaker / ICD	never on the plates: weight-only mode

Cost and scale

Home unit parts ₹4,319–4,486 (Monte Carlo of the itemised list; limit ₹5,000); ≈ ₹2,641–3,081 at 100 units; SMS ≈ ₹0.05 per patient-month. Uses the family's own chair, so nothing large to buy or store; fits ASHA-led programmes.

Honest limits

- Synthetic data shows the method works on modelled patterns, not clinical accuracy; lead time is bounded by the PS worsening window (3-10 days).
- Not yet built on a bench; power budget (datasheet values) needs firmware power saving for multi-day battery life; SpO_2 calibration generic; chest impedance is weak alone. Sit-to-stand time comes from the platform's force swing while he rises (the chair stays on it): a physics fit, 0.14 s median error on synthetic rises, not yet bench-tested.
- Never on anyone with a pacemaker or ICD; battery only while measuring. Code, data, tests: github.com/aadityaa0523/HridayMitra